

Kuramoto Oscillator Networks as a Distributed "Quantum-Like" Coherence Resource

John T. M. Campbell Department of Electronic Engineering Maynooth University, Ireland

Abstract

Coherence plays a central role in many quantum systems, from exciton transport in molecular aggregates to entanglement-based quantum information processing. However, maintaining entanglement as a functional resource in complex multimolecular environments remains challenging due to environmental decoherence. Recent work has therefore suggested that coherence itself may represent the operative resource governing energy transport and information propagation in complex systems.

In this work we propose that networks of Kuramoto phase oscillators provide a natural classical framework for modeling distributed coherence. Synchronization domains in oscillator networks are interpreted as analogues of exciton delocalization regions observed in molecular systems. We introduce distance-dependent power-law signaling as a coupling mechanism and interpret synchronization through a least-action perspective analogous to the path-integral formulation of quantum mechanics.

Experimentally, we analyze a network of optically interacting LEDs and reconstruct the underlying dynamics using Takens delay embeddings of measured intensity time series. In parallel, we simulate hierarchical Kuramoto networks consisting of interacting subnetworks (Alice, Charlie, Bob) with directional couplings and time delays. The results reveal mediated synchronization, lagging phase propagation, and cluster formation consistent with spectral graph embeddings.

These results suggest that Kuramoto oscillator networks can function as distributed coherence resources, providing an experimentally accessible platform for exploring synchronization-mediated information dynamics in complex systems and quantum-inspired network architectures.

1 Introduction

Coherence and entanglement are central concepts in quantum mechanics and quantum information science. Entanglement produces correlations between spatially separated systems that cannot be explained within classical physics and underpins many proposed quantum technologies [1, 2]. However, maintaining entanglement as a functional resource in complex many-body systems is often limited by environmental decoherence.

Recent work in molecular photophysics has emphasized the importance of coherent excitonic states in determining the efficiency of energy transport in molecular aggregates. In particular, studies of photosynthetic complexes have revealed that excitation energy may be delocalized across multiple molecules, producing quantum states that cannot be factored into independent local excitations.

For example, the exciton state of a molecular dimer consisting of molecules 1 and 2 may be written as

$$|J^+\rangle = \frac{1}{\sqrt{2}} (|1'2\rangle + |12'\rangle), \quad (1)$$

where the prime denotes the electronically excited molecule [5]. This state represents a coherent superposition of excitation localized on either molecule and is formally equivalent to a Bell-type entangled state.

In complex multimolecular systems, however, maintaining entanglement as a functional resource is difficult due to environmental interactions. It has therefore been proposed that coherence itself, rather than strict entanglement, may be the operative resource governing energy transport in such systems [5].

Kuramoto oscillator networks provide one of the most widely studied models of synchronization in nonlinear dynamical systems [3, 4]. Although originally developed to describe collective synchronization among chemical oscillators, the Kuramoto framework has been applied to a wide variety of complex systems including biological networks, laser arrays, and power grids.

In this work we explore the idea that synchronization in Kuramoto networks can be interpreted as a form of distributed coherence. Within this framework, clusters of synchronized oscillators correspond to coherence domains analogous to exciton delocalization regions in quantum systems.

2 Kuramoto Oscillator Networks

The Kuramoto model describes the dynamics of a population of coupled phase oscillators

$$\dot{\theta} * i = \omega_i + \sum_j K_{ij} \sin(\theta_j - \theta_i), \quad (2)$$

where θ_i represents the phase of oscillator i , ω_i its natural frequency, and K_{ij} the coupling strength between oscillators.

The collective coherence of the oscillator population is quantified using the complex order parameter

$$re^{i\Psi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}. \quad (3)$$

The magnitude r measures the degree of phase coherence across the network. When oscillators are completely unsynchronized their phases are uniformly distributed on the unit circle and $r \approx 0$. As synchronization develops, oscillator phases align and r approaches unity.

To quantify the evolution of synchronization we compute the time-dependent order parameter

$$r(t) = \left| \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)} \right|. \quad (4)$$

In the interpretation proposed here, $r(t)$ may be viewed as a measure of the effective coherence of the oscillator network.

3 Kuramoto–Exciton Analogy

The connection between excitonic coherence and Kuramoto synchronization can be made more explicit by comparing the exciton density matrix with the Kuramoto order parameter.

For a delocalized exciton state the coherence between two sites i and j is described by the off-diagonal density matrix element

$$\rho_{ij} = \langle i | \rho | j \rangle. \quad (5)$$

In a phase oscillator network the analogous quantity is the phase correlation between oscillators

$$C_{ij} = \langle e^{i(\theta_i - \theta_j)} \rangle. \quad (6)$$

Averaging these correlations across the network yields the Kuramoto order parameter

$$r = \left| \frac{1}{N} \sum_j e^{i\theta_j} \right|. \quad (7)$$

This formal similarity suggests that synchronization domains in Kuramoto networks can be interpreted as classical analogues of coherent exciton delocalization domains.

4 Distance-Dependent Signaling

Interactions in many physical systems decay with spatial separation. To model this effect we introduce a distance-dependent coupling kernel

$$\beta(r) = \beta_0 e^{-\gamma r^m}. \quad (8)$$

For optical systems the exponent $m = 2$ corresponds to the inverse square law of light propagation. The effective coupling between nodes becomes

$$K_{ij} = \beta_0 e^{-\gamma r_{ij}^2}. \quad (9)$$

Such spatially dependent interactions naturally lead to cluster formation, since nearby oscillators interact more strongly than distant ones.

5 Optical Oscillator Network Experiment

To explore these ideas experimentally we constructed a spatial network of optically interacting LEDs.

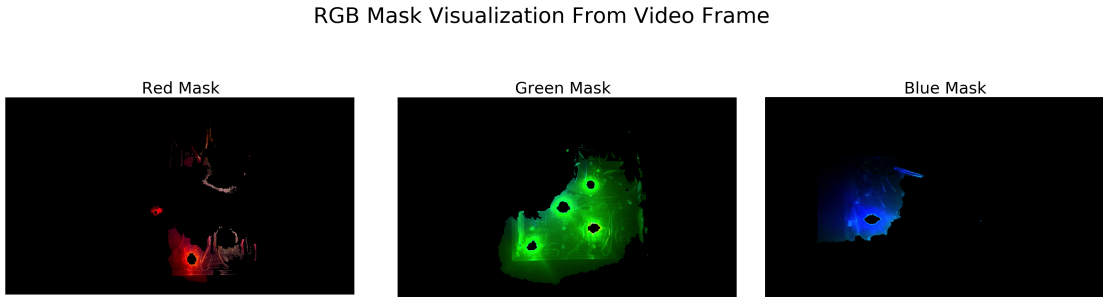


Figure 1: Snapshot of the LED oscillator network used in the experiment.

Frame-by-frame image processing was used to extract intensity signals for individual LEDs.

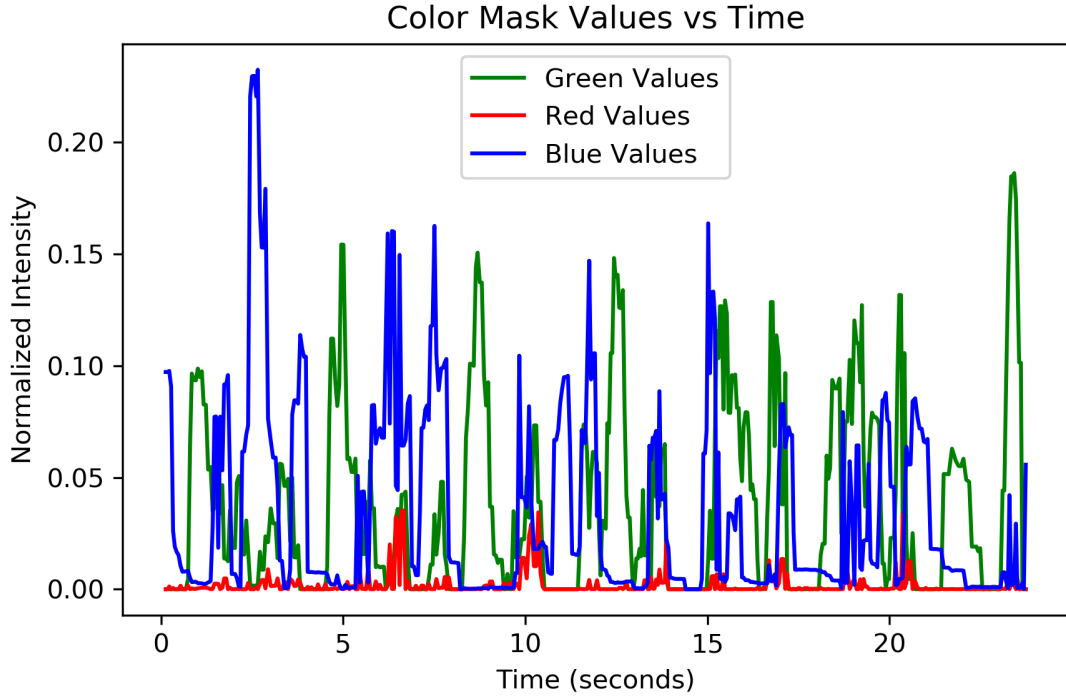


Figure 2: Intensity time series extracted from the red and blue LED channels.

The time series exhibit transitions between incoherent and partially synchronized regimes.

5.1 Takens Phase-Space Reconstruction

To visualize the underlying dynamics we reconstructed the system's phase space using Takens delay embedding

$$\mathbf{x}(t) = [x(t), x(t + \tau), x(t + 2\tau)]. \quad (10)$$

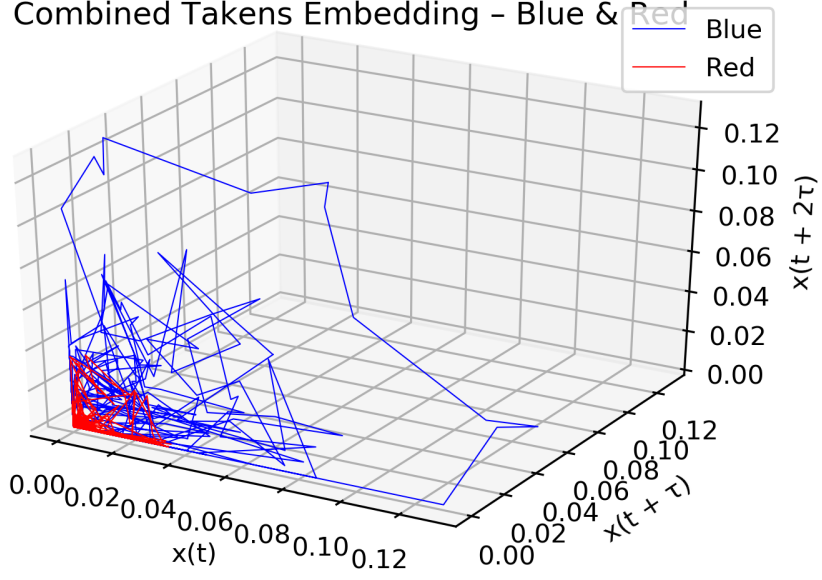


Figure 3: Reconstructed attractor for the LED oscillator network.

The reconstructed attractor reveals the geometry of the synchronization manifold governing the dynamics of the optical network.

6 Mediated Kuramoto Synchronization Networks

To investigate synchronization propagation we use a Quantum Kuramoto oscillator graph network simulation in Python (QKONGS) to simulate a hierarchical Kuramoto network consisting of three subsystems: Alice, Charlie, and Bob chained together where Charlie acts as the bridge network joining Alice and Bob.

$$\dot{\phi}_A = \omega_A \quad \dot{\phi} * C = \omega_C + K * AC \sin(\phi_A - \phi_C) \quad \dot{\phi} * B = \omega_B + K * CB \sin(\phi_C - \phi_B) \quad (11)$$

Here Charlie acts as an intermediate network mediating synchronization between Alice and Bob.

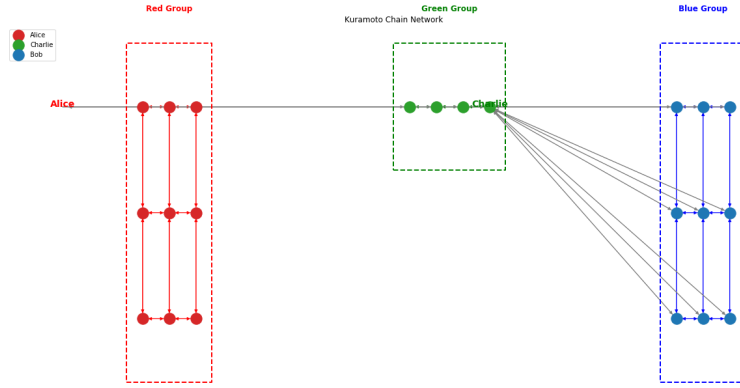


Figure 4: Topology of the Kuramoto chain simulation.

The time evolution of the Kuramoto order parameter is shown in Fig. 5. Initially the oscillators evolve independently, producing a low order parameter. As coupling effects propagate through the Charlie network, phase alignment develops and the global coherence increases.

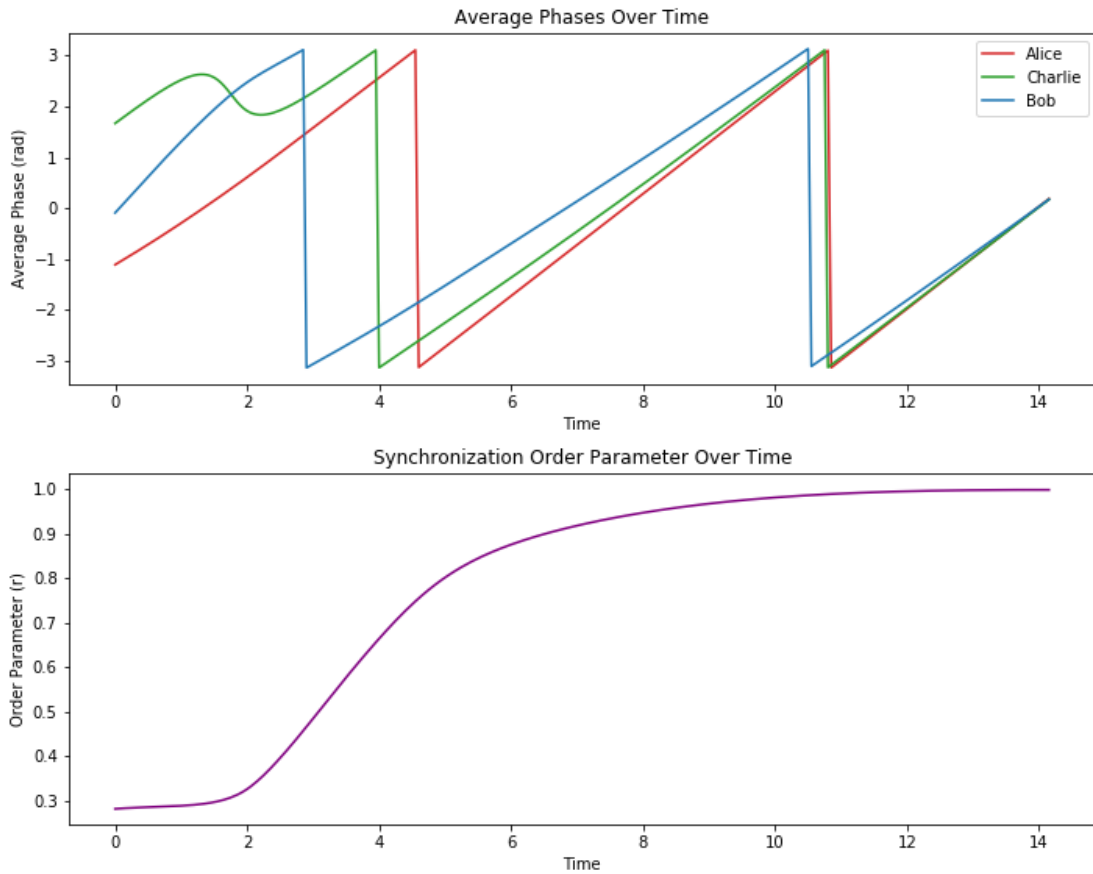


Figure 5: Evolution of network phases and Kuramoto order parameter $r(t)$.

7 Spectral Geometry of Coherence Networks

The structural organization of oscillator networks can be analyzed using spectral graph methods. The Laplacian matrix

$$L = D - A \quad (12)$$

encodes the connectivity of the system.

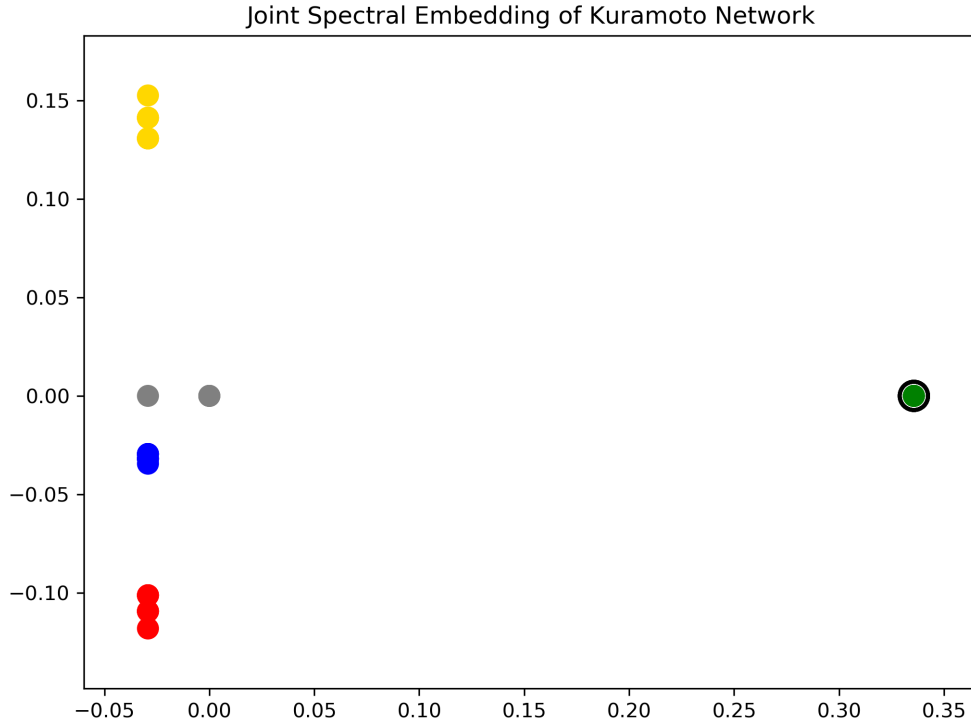


Figure 6: Spectral embedding of the four-cluster Kuramoto network.

Clusters appear as distinct regions in spectral space, with the Charlie nodes occupying a central position.

Figure 7: Fiedler-sorted adjacency matrix revealing network community structure.

These spectral structures reflect the emergence of coherence domains within the oscillator network.

8 Conclusion

We have proposed a framework that interprets synchronization in Kuramoto oscillator networks as a form of distributed coherence. By combining optical experiments with numerical simulations we show that oscillator networks can develop coherent dynamical structures analogous to exciton delocalization domains in molecular systems.

This perspective suggests that classical oscillator networks may serve as experimentally accessible platforms for exploring coherence dynamics relevant to complex quantum systems, photonic networks, and synchronization-based information processing architectures.

References

- [1] M. Nielsen and I. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (2010).
- [2] R. Horodecki et al., Quantum entanglement, *Rev. Mod. Phys.* 81, 865 (2009).
- [3] Y. Kuramoto, Self-entrainment of a population of coupled oscillators, *Lecture Notes in Physics* (1975).
- [4] J. Acebrón et al., The Kuramoto model: a simple paradigm for synchronization, *Rev. Mod. Phys.* 77, 137 (2005).
- [5] G. D. Scholes, Limits of exciton delocalization in molecular aggregates, *Royal Society Open Science* 6 (2019).

9 Appendix

9.1 Code

QKONGs: <https://github.com/MuonRay/Quantum-Kuramoto-Oscillator-Network-Graph-Simulation>